

Andrew Brettin

📍 New York, NY ✉ brettin@cims.nyu.edu 🌐 [Website](#) 🐙 [GitHub](#) 🔗 [LinkedIn](#)

Climate Data Scientist with a PhD in Atmosphere-Ocean Science & Mathematics and 5+ years of experience in predictive modeling and uncertainty quantification. Expert in building machine learning algorithms for assessing weather and climate risk. Proven ability to manage TB-scale machine learning pipelines and communicate insights to stakeholders.

Relevant Experience

Courant Institute of Mathematical Sciences, New York University <i>Postdoctoral Research Associate</i> <i>Graduate Research Assistant</i> Topics: Machine learning for quantifying climate risks and sea level predictability.	New York, NY May 2025–September 2025 September 2019–May 2025
Mathematics and Climate Research Network, University of Minnesota <i>Research Assistant</i> Topics: Mathematical modeling of biodiversity in ecological systems.	Minneapolis, MN September 2017–May 2019
Center for Discrete Mathematics and Computer Science, Rutgers University <i>NSF Summer Research Intern</i> Topics: Stochastic models of cancer growth. Reaction-diffusion equations, Monte Carlo simulation.	New Brunswick, NJ May 2018–July 2018

Education

Ph.D., Courant Institute, NYU. Atmosphere-Ocean Science & Mathematics. Advisor: Laure Zanna	May 2025
B.Sc., University of Minnesota. Mathematics, <i>summa cum laude with high distinction</i> (GPA 3.92)	May 2019
<i>Awards and distinctions:</i> VoLo Fellowship (2020) • Henry M. MacCracken Fellowship (2019) • Hans H. Dalaker Scholarship (2018) • NSF REU Fellow (2018 & 2017) • Gold Scholar Award (2015)	

Research Projects

Estimating temperature and precipitation extremes with quantile neural networks - Designed and benchmarked a novel machine learning framework for uncertainty quantification, outperforming industry-standard baselines across 1,500+ weather stations - Engineered multi-source features by integrating gridded climate data (Google Cloud ERA5) with in-situ observations (NOAA GHCN data), strategically combining reanalysis products with local measurements to improve predictive performance - Validated model robustness using probabilistic evaluation metrics (CRPS, calibration), minimizing forecast bias and improving confidence interval reliability <i>Outcomes:</i> [preprint] [code]	2025
Learning efficient forecasts for regional sea surface height dynamics - Customized a deep learning framework (convolutional autoencoders) with nonlinear dynamical systems theory (Koopman operators) to transform high-dimensional, complex spatiotemporal data into a tractable, linear latent representation - Implemented Distributed Data Parallelism (DDP) in Pytorch to efficiently scale training over 200+ GB of spatiotemporal data on multiple GPUs - Demonstrated superior predictive performance over stochastic Linear Inverse Models (LIM), improving forecasting skill by 5–10% using latent representations of similar complexity <i>Outcomes:</i> [Article, Geophysical Research Letters] [code] [presentation]	2024–2025
Identifying sources of sea level predictability with mean-variance networks - Architected scalable ETL pipelines to process TB-scale spatiotemporal climate datasets, leveraging Dask and Xarray for parallel processing on high-performance computing (HPC) systems - Automated end-to-end training, optimization, and evaluation of an ensemble of 6,500+ neural networks	2023–2024

- Identified physical drivers of sea level predictability using Explainable AI (XAI) techniques

Outcomes: [\[Article, *Artificial Intelligence for the Earth Systems*\]](#) [\[code\]](#) [\[conference presentation\]](#)

Exploring nonstationarity of coastal sea level distributions

2021–2022

- Developed a non-stationary statistical framework for quantifying shifting probability distributions in tide gauge observations, enabling more robust risk assessment of extreme events under shifting climate states

Outcomes: [\[Article, *Environmental Data Science*\]](#) [\[code\]](#) [\[presentation\]](#)

Technical Skills

Languages	Python (numpy, pandas, scipy, scikit-learn, matplotlib) • Bash • Julia • R • SQL • C++
MLOps	Linux • VCS (git/GitHub) • Virtual environments (conda) • Containerization (Docker/Singularity) • Pytorch/TensorFlow • Experiment tracking (Lightning AI/Weights & Biases) • Unit testing (pytest)
Computation	HPC clusters (SLURM/PBS) • Cloud platforms (GCP) • Distributed/parallel computing • GPU acceleration (CUDA/DDP)
Geospatial	Raster data strategies (xarray, Dask, ESMF regridding) • Temporal data strategies (cftime, flox) • File formats (NetCDF/HDF5/zarr/GRIB) • Visualization (cartopy/HoloViews)
Stats/Math	Uncertainty quantification • Extreme value analysis • Quantile regression • Bayesian inference • PCA • Unsupervised learning • Time-series forecasting • Numerical methods
ML	CNNs • Autoencoders • Random forests • Gaussian processes • U-nets • Diffusion models

Publications

1. **Brettin, A.** & Zanna, L. Estimation of temperature and precipitation uncertainties using quantile neural networks (preprint). 2026. arXiv: [2601.17243](#) [\[physics.aos-ph\]](#).
2. **Brettin, A.** *Data-Driven Approaches for Predictability and Uncertainty Quantification of Sea Level Variability*. [3223742845](#). PhD thesis (New York University, 2025).
3. **Brettin, A.**, Zanna, L. & Barnes, E. Learning Propagators for Sea Surface Height Forecasts Using Koopman Autoencoders. *Geophys. Res. Lett.* **52**, [e2024GL112835](#) (2025).
4. **Brettin, A.**, Zanna, L. & Barnes, E. Uncertainty-Permitting Machine Learning Reveals Sources of Dynamic Sea Level Predictability across Daily to Seasonal Time Scales. *Artif. Intell. Earth Syst.* **4**, [250014](#) (2025).
5. Falasca, F., **Brettin, A.**, Zanna, L., Griffies, S. M., Yin, J. & Zhao, M. Exploring the nonstationarity of coastal sea level probability distributions. *Env. Data Sci.* **2**, [e16](#) (2023).
6. Meyer, K., Broda, J., **Brettin, A.**, Sánchez Muñoz, M., Gorman, S., Isbell, F., Hobbie, S. E., Zeeman, M. L. & McGehee, R. Nitrogen-induced hysteresis in grassland biodiversity: a theoretical test of litter-mediated mechanisms. *Am. Nat.* **201**, [E153–E167](#) (2023).
7. **Brettin, A.**, Rossi-Goldthorpe, R., Weishaar, K. & Erovenko, I. V. Ebola could be eradicated through voluntary vaccination. *Royal Society Open Science* **5**. [10.1098/rsos.171591](#), [171591](#) (2018).

Additional Activities

- **NASA/JPL Summer School on Satellite Observations and Climate Models**, Caltech, Pasadena, CA 2023
- **LEAP Momentum Bootcamp on Climate Data Science**, Columbia University, New York, NY 2022
- **Volunteer tutor, math grades 5–8**, Common Denominator, New York, NY 2021–2022
- **Vice President, Courant Student Organization**, New York University, New York, NY 2021–2022
- **Project mentor, Undergraduate Research Program in Data Science** 2021
NYU Center for Data Science (collaboration with the National Society of Black Physicists), New York, NY
- **Reviewer**, *Geophysical Research Letters* (2025–) and *Journal of Geophysical Research: Machine Learning and Computation* (2025–)